REFLECTION JOURNAL – **Module 08: Bias–Variance Tradeoff**

ITAI 1371

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Date: October 24, 2026

Learning Experience - In this lab, I used Python and Scikit-learn to generate a noisy sine-wave dataset and trained polynomial regression models with degrees 1, 4, and 15. The graphs helped me understand how model complexity affects prediction performance.

Observations from My Graphs - In the first graph, the degree 1 model produced a nearly straight, slightly downward-sloping blue line that failed to capture the wave pattern, showing underfitting. The degree 4 orange curve followed the general wave shape more closely, providing a reasonable fit. The degree 15 green curve followed more fluctuations in the data, indicating potential overfitting.

In the learning curves, the degree 1 training score stayed close to zero, while its validation score was very negative with smaller training samples and improved as the sample size increased. For degree 15, some validation scores became extremely negative, with the graph displaying a scale around (10^{18}). This suggested poor generalization and possible numerical instability.

Challenges and Solutions - I encountered syntax, indentation, and NameError issues while writing the code. I corrected the line breaks, indentation, and function execution order, then successfully generated the graphs. This improved my debugging skills and understanding of Python functions in Google Colab.

Conclusion - This lab taught me that increasing model complexity does not always improve performance. I learned to identify underfitting and potential overfitting from graphs and to use training and validation scores to evaluate model performance and generalization.

L08 – Reflection – Alexander DeBusk

During this lab, I learned about the difference between underfitting and overfitting and how they affect a machine learning model. One of the main things I worked on was polynomial regression, where I trained models with different degrees to see how well they fit a dataset containing a sine wave with random noise. I learned that a degree 1 model was too simple to capture the pattern, while a degree 15 model was too complex and started following the noise. The degree 4 model provided a better balance between the two.

One challange I encountered was making sure the code was formatted correctly and that each model was trained and plotted properly. I also had to understand how the learning curve function worked and how it compared training scores with cross-validation scores. At first, it was a little confusing to understand what the graphs were showing, but comparing the two models helped me understand the difference more clearly.

Another important insight I gained was that a model performing well on training data does not always mean it will perform well on new data. This is why cross-validation is useful when evaluating machine learning models. I also learned that adding more training data does not always fix every problem, especially when the model itself is too simple. These techniques are usefull in generative AI because models need to learn patterns from training data without simply memorizing examples. Finding the right balance can help a model make better predictions and produce more reliable results when working with unsean data.

Reflection

In this lab, I learned how model complexity affects machine learning performance by comparing polynomial models with degrees 1, 4, and 15. This helped me understand the difference between underfitting, overfitting, and a good fit. Before this lab, I thought a more complex model might always give better results. However, I learned that a degree 15 model can fit the training data very closely but still perform poorly on new data because it also learns noise. In comparison, the Degree 1 model is too simple to follow the main pattern, while the Degree 4 model gives a better balance. I learned that underfitting happens when a model is too simple to learn the patterns in the data. Overfitting happens when a model learns the training data too closely and does not work well on new data. Finding a good balance between these two problems is important for building a useful model. Learning curves helped me compare training scores and cross validation scores. I noticed that an underfitting model has low scores for both, while an overfitting model has a high training score but a much lower cross validation score. This helped me understand how to identify problems and evaluate model performance more clearly.